



TransOrg's AgentIQ Datathon:

From Messy Data to Agentic Insights

Clean It. Analyze It. Visualize It. Ask It.

An Enterprise-Grade Data Engineering, Analytics & AgenticAI Challenge



Important Details at a Glance

- 🎯 Problem Statement Release: 11th Sept, 2026 at 11:00 AM
- 📅 Individual Registration Deadline: 12th Sept, 2026 at 11:59 PM
- ⌚ Submission Deadline: 15th Sept, 2026 at 11:59 PM
- 💻 Event Mode: 100% Online (*Grand Finale for Top 5 teams will be Offline at Shanti Devi Mittal Auditorium on 18th Sept, 2026*)
- 🏆 Prize Pool: ₹30,000 + Certificates for all participants
- 👥 Team Size: 3 to 5 Members (*Cross-functional teams recommended! Solo allowed but not recommended*)
- 🎓 Eligibility: Being an LPU student is mandatory, that's it. (*If you can play with Python, feel free to register.*)
- 🔗 Register Here: [Individual Registration Form](#)
- 💬 Join WhatsApp Group [Mandatory]: [WhatsApp group to join](#)



What is this Datathon?

Forget clean, pre-packaged Kaggle datasets. In the real world, data is messy, incomplete, and chaotic.

Inspired by enterprise analytics consulting (like TransOrg Analytics), this datathon challenges you to act as a "Data Squad" for a hypothetical client. You will be handed a highly imperfect, "messy" dataset. Your mission is to rescue the data, build a governed analytics pipeline, create an executive dashboard, and—if you're feeling

ambitious—build a Bonus AI Agent that can answer business questions by automatically generating graphs!

The 4-Layer Challenge

1. **Data Rescue (Core):** Clean the messy data, handle duplicates, fix broken relationships, and standardize formats.
2. **Analytics Layer (Core):** Define business metrics (Revenue, Churn, ROI) and build a structured, query-ready data model.
3. **Executive Dashboard (Core):** Build an interactive dashboard (PowerBI, Tableau, Streamlit, etc.) that tells a compelling business story.
4.  **BONUS: Agentic Graph AI:** Build a chatbot/agent that takes natural language questions (e.g., *"Compare Q3 sales by region"*) and dynamically generates the correct chart (Bar, Line, Scatter) to answer it.

Event Timeline

Milestone	Date & Time
Registration Opens	2nd Sept, 2026
Kickoff & Problem Statement Release	11th Sept, 2026 @ 11:00 AM
Datathon Execution Window	11th Sept to 15th Sept, 2026
Final Submission Deadline	15th Sept, 2026 @ 11:59 PM
Top 5 Finalists Announced	17th Sept, 2026
Grand Finale & Presentations (Offline)	18th Sept, 2026 @ 09:00 AM

The 4 Problem Statements (Choose One Track)

Your team must select ONE of the following industry tracks. You will be provided with the raw, messy datasets specific to your chosen track.

Track 1: FinTech & BFSI - "UPI Fraud Ring & Merchant Analytics"

- The Business Problem: A National Payments Authority needs to analyze micro-transaction graphs to identify circular money-laundering rings, synthetic identity fraud, and compromised merchant accounts.
- The "Messy" Data: High-velocity logs with missing UTR numbers, mismatched PAN/Aadhaar formats, OCR errors, and currency symbols embedded in numeric columns.
- Bonus Agent Query: *"Which merchant category has the highest chargeback-to-transaction ratio this quarter?"*

Track 2: Cybersecurity - "Zero-Trust Telemetry & Insider Threat Logs"

- The Business Problem: A cyber-command center needs to detect compromised accounts and insider threats by analyzing massive, noisy network logs and IAM audit trails.
- The "Messy" Data: Millions of firewall log rows with truncated IPs, JSON audit trails with mismatched session IDs, and unstructured antivirus alert text.
- Bonus Agent Query: *"Show the trend of failed login attempts by department over the last 7 days."*

Track 3: AgriTech - "Mandi-to-Market Supply Chain Optimizer"

- The Business Problem: A State Agriculture Board needs a dashboard to monitor daily crop arrivals at local Mandis, track prices against the MSP, and correlate with weather data.
- The "Messy" Data: Crop names mixed in English and Hindi, quantities mixing units (Quintals vs. KGs), and weather sensor logs with timezone mismatches.
- Bonus Agent Query: *"Plot the daily arrival trend of Wheat in Amritsar mandi vs MSP for the last 30 days."*

Track 4: 🎓 Education & EdTech - "Student Retention & Welfare Efficacy Tracker"

- The Business Problem: The State Education Dept needs to track the correlation between the Mid-Day Meal scheme, school infrastructure, and student dropout risks.
- The "Messy" Data: Proxy attendance anomalies, grain procurement logs with unit mismatches, and infrastructure surveys with Boolean columns filled with "Yes", "Y", "1", "Hai", "Nahi".
- Bonus Agent Query: *"Compare average test scores between schools with and without functional electricity."*



The "Knockout" Evaluation Rubric (How You Will Be Judged)

Because we expect a massive number of registrations, we use a **"Stage-Gate Knockout"** System. Evaluators will check your submission against "Gates". If you fail a Gate, your evaluation stops there.

Pro-Tip: Write an incredibly clear `README.md` file. Evaluators will read your `README` before looking at your code!



Gate 1: The Compliance & Sanity Check (Pass/Fail) (Score: 0 to 10)

Before we look at your code, we check if you followed basic professional standards.

- Is the GitHub repository public and accessible?
- Is there a well-formatted [README.md](#) explaining how to run the project?
- Is a Data Dictionary provided (explaining what each column means)?
- Is there proof of data cleaning (e.g., a script showing raw row count vs. cleaned row count)?
- 🛑 KNOCKOUT CONDITION: If any of these 4 items are missing, the team is eliminated from the competition. No further evaluation will occur.
- ✨ 10 BONUS pts, if [README.md](#) is not AI-generated and developed by self.

📖 Gate 2: The Data Engineering & Rescue Check (Score: 0 to 30)

Evaluators will look at your Python/SQL scripts to see how you handled the "messy" data.

- 10 pts: Handling missing values and duplicates (Did you intelligently impute/fix, or did you just lazily drop 50% of the rows?).
- 10 pts: Standardization (Did you fix the mixed date formats, unit mismatches, and multilingual text?).
- 10 pts: Reproducibility (Can the evaluator run your script/notebook from top to bottom without errors?).
- 🛑 KNOCKOUT CONDITION: If your data is still fundamentally broken (e.g., dates are still unparseable strings, or relationships between tables are completely ignored), you score below 10 and are eliminated.
- 🌟 10 BONUS pts, if your notebook has appropriate comments explaining your decision.

📖 Gate 3: The Dashboard & Business Value Check (Score: 0 to 40)

Evaluators will interact with your Dashboard (PowerBI, Tableau, Streamlit, etc.).

- 15 pts: Interactivity & UX (Are there filters? Is the layout clean and professional?).
- 15 pts: Core KPIs (Did you accurately calculate the main business metrics like Revenue, Churn, Risk Score?).
- 10 pts: Storytelling (Does the dashboard actually answer the core business problem, or is it just a dump of random pie charts?).
- 🛑 KNOCKOUT CONDITION: If the dashboard is just static screenshots with no interactivity, or if it fails to show the required core metrics, you are eliminated.
- 🌟 10 BONUS pts, for utilizing innovative approaches for the dashboard.

📖 Gate 4: The Excellence & AI Bonus Check (Score: 0 to 30 + 30 Bonus)

- 15 pts: Code Elegance & Architecture (Use of modular functions, SQL views, clean pipeline structure).

- 15 pts: Advanced Insights (Did you find hidden trends? Did you use forecasting or clustering?).
- 🌟 30 BONUS pts (Agentic Graph AI):
 - Does the agent understand natural language? (10 pts)
 - Does it correctly select and render the right chart type (Bar vs Line vs Scatter)? (10 pts)
 - Does it provide a text summary alongside the graph? (10 pts)



Step-by-Step Guide: How to Approach the Datathon

To win, treat this like a real corporate consulting sprint. Follow this 4-day execution plan:

- **Day 1 (Sept 11): Data Rescue & EDA.** Don't jump into dashboards. Spend the first 24 hours writing Python scripts to clean the data. Handle the missing values, fix the date formats, standardize the text, and remove duplicates. Document every cleaning step.
- **Day 2 (Sept 12): The Analytics Layer.** Define your metrics. What is the exact mathematical formula for "Churn" or "Fraud Risk" in your dataset? Build a clean, structured SQL database or Pandas DataFrames that are ready for querying.
- **Day 3 (Sept 13-14): Dashboarding & Storytelling.** Connect your clean data to PowerBI, Tableau, or Streamlit. Build KPI cards at the top, trend lines in the middle, and detailed tables at the bottom. Ensure your dashboard answers the core business problem.
- **Day 4 (Sept 15): The Bonus Agent & Polish.** Build your Text-to-Chart AI agent. Test it with edge cases. Record your 3-5 minute demo video, write a clean `README.md` for your GitHub repo, and submit everything before 11:59 PM!



Submission Guidelines & Rules

Where to submit: All code and data dictionaries must be pushed to a public GitHub Repository. A link will be shared in the WhatsApp group to submit your dashboard link [a live deployed link is mandatory] , final presentation [PPT needs to be in PDF only], and GitHub repo.

- One Submission Per Team: The contest will allow only one account per participant. Submissions from multiple accounts will lead to disqualification.
 - No Private Sharing: Privately sharing code or data outside of your team is not permitted. It's okay to share code if made available to *all* participants on the official forums/WhatsApp group.
 - Disqualification: If any of the details entered are found incorrect, TransOrg Analytics reserves the right to disqualify any participant.
 - Plagiarism: You are free to use any tool or resource, but any peer copy or plagiarism is strictly prohibited.
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? Frequently Asked Questions (FAQs)



General & Eligibility

Q: I can see the recommended team size is 3-5, but there is no option for team registration in the form.

A: At the first stage, each participant needs to register individually. During the objective release, another form will be shared for updating the team details. Join the WhatsApp group to follow the updates.

Q: Can I participate as a single-member team?

A: Yes, solo participation is technically allowed. However, we highly recommend bringing in at least 3 members from different backgrounds so that it's easier to complete the objective within the given timeline.

Q: Do I need to be an expert in AI/Machine Learning to participate?

A: No! The core of this event is Data Engineering, Data Cleaning, and Business Intelligence. The AI Agent is strictly a bonus layer. Teams with excellent data pipelines and dashboards can easily win top prizes without building an AI agent.

Q: What's the event flow throughout the whole datathon?

A: Register -> Join WhatsApp -> Problem Statement Released (Sept 11) -> Develop Solution -> Submit GitHub/Dashboard links (Sept 15) -> Top 5 Announced (Sept 17) -> Grand Finale Presentations & Prize Distribution (Sept 18).

Q: Can 1st and 2nd-year students participate?

A: Yes! This is a great opportunity to learn enterprise data practices early. You can team up with seniors to balance the skill sets.

Q: Is there any registration fee?

A: No, the event is completely free for all LPU students.

Q: Will there be any certificates for the participants?

A: Yes, all participants will receive a certificate of participation.

Q: Can you share the cash prize details?

A:

- 1st Prize: ₹11,000
- 2nd Prize: ₹7,000
- 3rd Prize: ₹4,000
- Best AI Agent: ₹3,000 (special category)
- Best Dashboard: ₹2,500 (special category)
- Best Data Pipeline: ₹2,500 (special category)

Technical & Tools

Q: What tools and tech stack are we allowed to use?

A: You have complete freedom! The more innovative, the better.

- *Data Engineering*: Python, Pandas, SQL, DuckDB, PySpark, dbt.
- *Dashboarding*: Power BI, Tableau, Streamlit, Plotly Dash, Looker.
- *Bonus AI*: OpenAI API, LlamaIndex, LangChain, HuggingFace, Plotly, etc.

Q: Will we have to pay for Cloud or LLM API credits?

A: No. You are expected to use free tiers, open-source local models (like Llama 3 via Ollama), or student developer packs. Do not spend your own money on enterprise cloud tiers.

Q: How "messy" is the dataset going to be?

A: Very. Expect missing values, mismatched IDs, wrong data types, duplicate records, and inconsistent text. You will need to write scripts to clean and validate it.

Q: What does "Graph-First AI Agent" mean?

A: If a user asks *"What is the total revenue?"*, a text answer is fine. But if they ask *"Show me the revenue trend over the last 6 months"*, the agent shouldn't just spit out numbers; it should generate and display a line chart.

Need Help? Contact the Organizers!

Got a question that isn't answered here? Reach out to the core organizing team:

- Student Coordinator: Harsh Bhardwaj | 9491091197
- Faculty Coordinator: Gurpreet Kaur | 6284363866
- Faculty Coordinator: Priyanshu Shukla | 7389731173

👉 Don't wait! Secure your team's spot and start brainstorming your tech stack today.